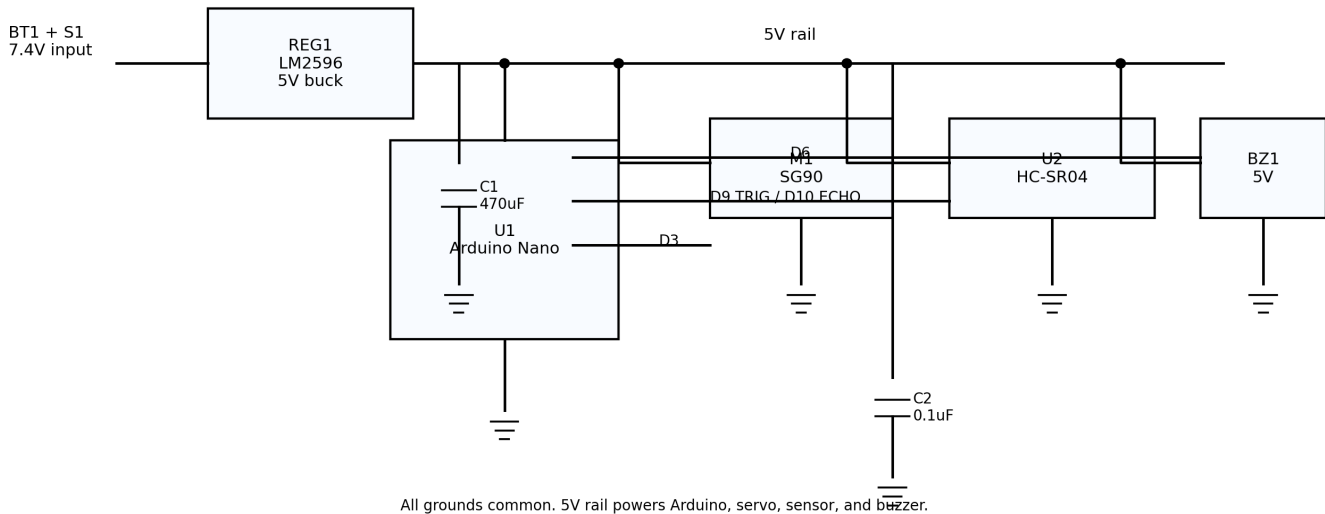


Too-Close Scanning Alarm Circuit Package

Battery-powered scanning proximity alarm that sweeps an ultrasonic sensor and sounds a buzzer when an object is detected closer than approximately 1 meter.

Schematic



Schematic QA pass: power, grounds, functional blocks, and firmware pin labels are internally consistent.

Bill of Materials

Ref	Item	Specification	Qty	Unit	Ext.	Supplier
BT1	2x18650 battery holder with cells	2-cell Li-ion pack, 7.4V nominal	1	\$12.00	\$12.00	Generic / Amazon
S1	Mini SPST slide switch	Power switch, $\geq 1A$ @ 12V	1	\$0.80	\$0.80	Adafruit / Amazon
REG1	LM2596 buck converter module	7-24V in, 5V out adjustable, $\geq 1A$	1	\$2.50	\$2.50	Amazon / eBay
U1	Arduino Nano compatible	ATmega328P development board	1	\$6.50	\$6.50	Amazon / SparkFun
U2	HC-SR04 ultrasonic sensor module	2cm-400cm distance sensor	1	\$2.20	\$2.20	Amazon / Adafruit
M1	SG90 micro servo	5V 9g positional servo	1	\$3.20	\$3.20	Amazon / TowerPro
BZ1	Active buzzer module	5V self-oscillating buzzer	1	\$1.50	\$1.50	Amazon
C1	Electrolytic capacitor	470uF 10V radial	1	\$0.25	\$0.25	DigiKey
C2	Ceramic capacitor	0.1uF 50V X7R	1	\$0.05	\$0.05	DigiKey
J1	Hook-up wire / jumper set	22 AWG assorted	1	\$2.00	\$2.00	Amazon
HW1	Sensor mounting bracket	HC-SR04 to SG90 bracket	1	\$1.50	\$1.50	Amazon / 3D print

Estimated total cost: **\$32.50 USD**

Firmware file: `too_close_alarm.ino`

Detailed purchase links are included in `bom.csv` and `bom.md`.

Assumptions / limitations

- Scanning head uses SG90 + HC-SR04.
- Power assumes a 2-cell Li-ion source with a 5V buck regulator.

- Ultrasonic measurements can degrade on soft or angled surfaces.
- BOM pricing is approximate and was not live-verified.